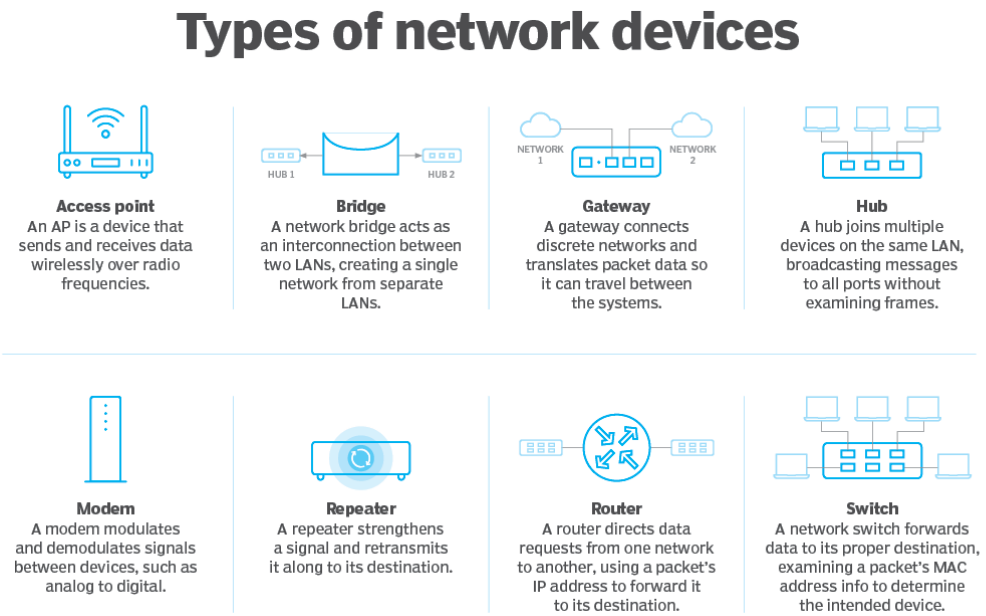
Intro to Networking Devices-01

Networking devices are hardware components used to connect computers, servers, and other devices to share data and resources. Common devices include:

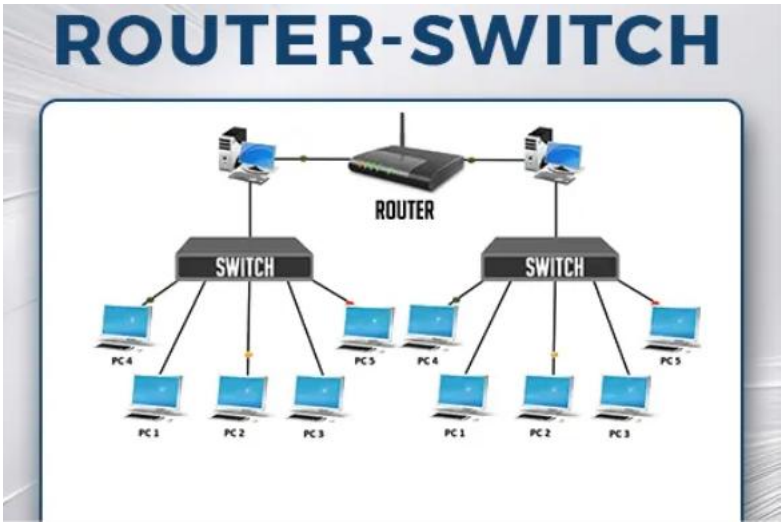


1.Switch – Connects devices in a local network and forwards data based on MAC addresses.

Types of Switches

- **Unmanaged Switch**: Basic, plug-and-play, no configuration.
- **Managed Switch**: Offers VLANs, monitoring, and security features.
- **Layer 3 Switch**: Works like a router, supports routing between VLANs.

2.Router – Connects multiple networks and forwards data based on IP addresses.



Functions

- Provides Internet access.
- Performs Network Address Translation (NAT).
- Acts as a DHCP server.
- Supports routing protocols (OSPF, RIP, BGP).

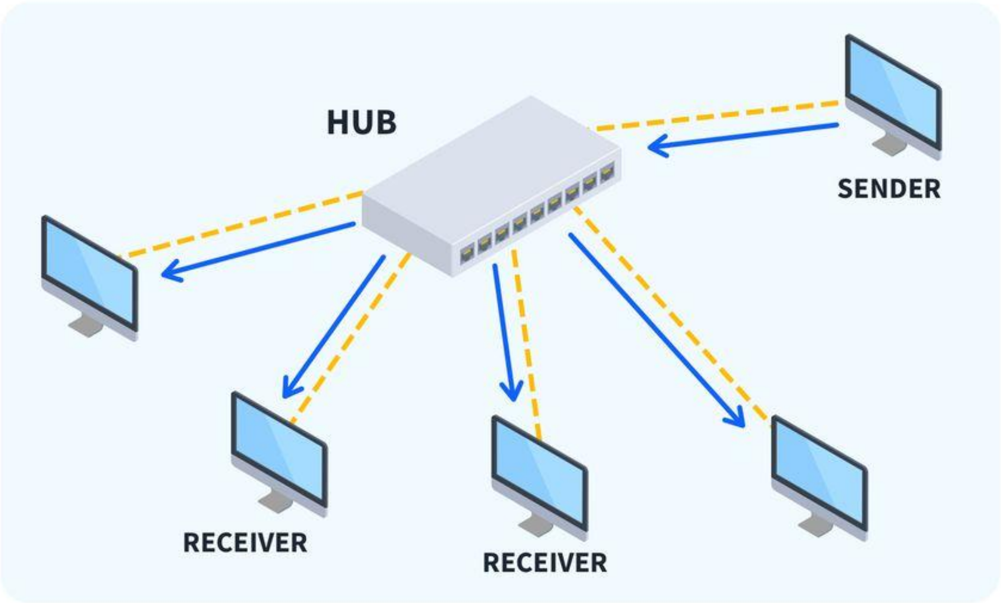
IPv4 and IPv6

- IPv4 – 32-bit addressing, written as four decimal numbers (e.g., 192.168.1.1).
- IPv6 – 128-bit addressing, written in hexadecimal (e.g., 2001:0db8:85a3::8a2e:0370:7334).
- Static IP – Manually assigned, does not change.
- Dynamic IP – Assigned automatically by DHCP.

Example

Your home Wi-Fi router connects your devices to the Internet.

3.Hub – A basic device that broadcasts data to all connected devices.



Types of Hubs

- Passive Hub – Only connects devices, no signal boosting.
- Active Hub – Amplifies the signal before forwarding.
- Intelligent Hub – Provides basic management features like monitoring and filtering.

Example
In a small office, a hub connects 6 computers in a single LAN, sharing data across all devices.

4.Access Point – Provides wireless connectivity to devices.

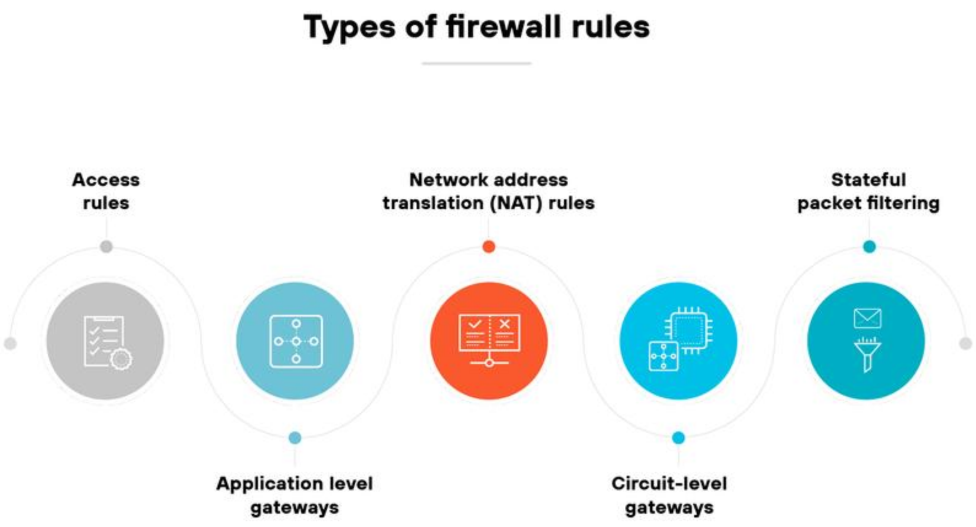
Types of Access Points

- **Standalone AP** – Works independently.
- **Controller-based AP** – Managed centrally.
- **Cloud-managed AP** – Controlled via cloud platforms.

Example

In a university campus, multiple APs provide wireless Internet across buildings.

5.Firewall – Controls and filters network traffic for security.



Types of Firewalls

- **Packet Filtering Firewall** – filters based on IP, port, protocol.
- **Stateful Inspection Firewall** – tracks active connections.
- **Next-Generation Firewall (NGFW)** – includes deep packet inspection, intrusion detection, and prevention.

Functions

- Blocks unauthorized access.
- Prevents cyber-attacks.
- Enforces security policies.

Example

An enterprise firewall may block all traffic except HTTP (port 80) and HTTPS (port 443).

6. Intranet

An intranet is a **private network** used inside an organization. It works like the internet but is limited to employees or members of that organization.

Key points:

- Accessible only within the company or through secure VPN.
- Hosts internal websites, apps, and services.
- Used for sharing documents, policies, announcements, and collaboration tools.

- Improves communication between employees.

Overview of LAN and WAN Technologies

LAN (Local Area Network) – Covers a small area like an office, home, or campus.

WAN (Wide Area Network) – Covers large geographical areas; the internet is the largest WAN.

Crossover Cable – A special Ethernet cable used for direct connection between two computers without a switch.

Peer-to-Peer Network – Each device can share resources without a dedicated server.